Simulated Annealing (SA) — Design Choices, Calibration, and TSP Case Study

IWRI Workshop day 2

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Agenda

1. SA in a Nutshell
2. Reference Pseudocode
3. Temperature Schedule
4. Further Design Choices
5. TSP Case Study

Roadmap

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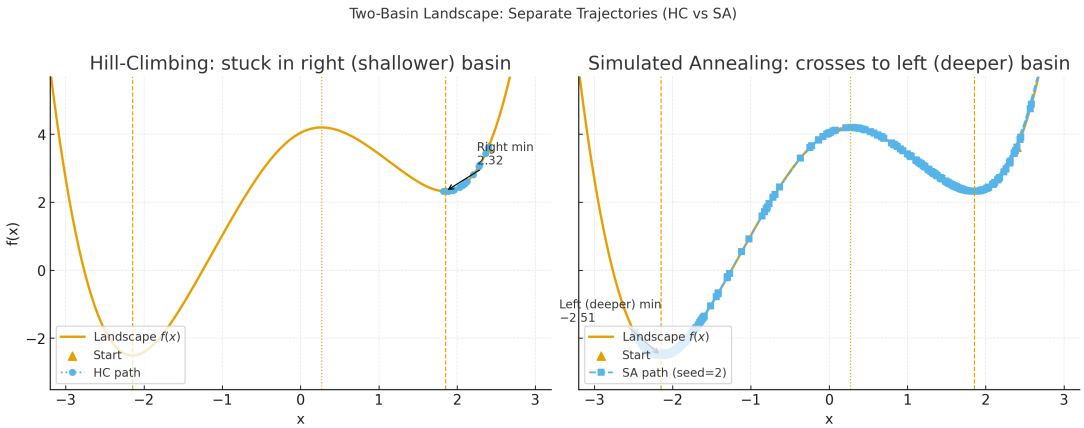
HC vs SA on a toy landscape



Same start and neighbourhood; different acceptance: HC gets stuck (right), SA crosses to deeper basin (left).

Starting point: Local Search + Metropolis

We start from a standard local search loop, and add the Metropolis criterion to occasionally accept uphill moves:

- Generate a neighbour $x' \in N(x)$ and compute $\Delta = f(x') - f(x)$.
- **If** $\Delta \leq 0$ **then** accept (descent step).
- **Else** accept with probability $e^{-\Delta/T}$ (exploration).
- As $T \downarrow 0$, this reduces to greedy local search.

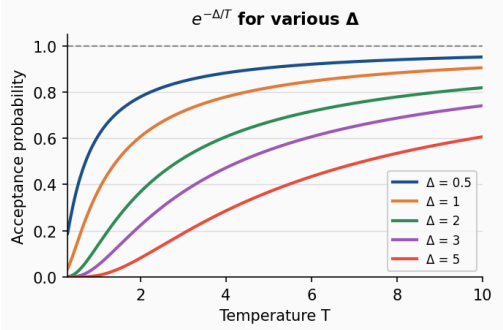
Key idea: SA is LS + a temperature-controlled chance to cross small uphill barriers.

Metropolis acceptance (core idea)

Given candidate $x \to x'$ with cost change $\Delta = f(x') - f(x)$ at temperature $T > 0$:

$$\Pr(\text{accept}) = \min\{1,\ e^{-\Delta/T}\}.$$

- $\Delta < 0$: always accept (improvement).
- $\Delta > 0$: accept with probability $e^{-\Delta/T}$ (controlled exploration).
- High $T \Rightarrow$ explores freely; $T \downarrow 0 \Rightarrow$ SA reduces to greedy descent.

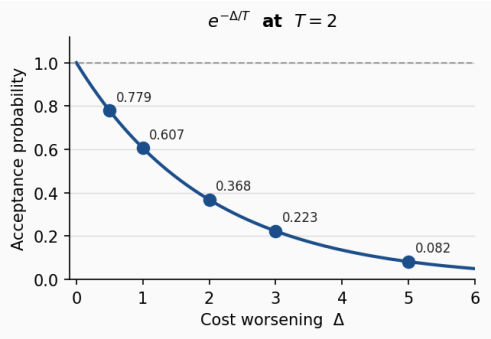


$e^{-\Delta/T}$ plotted for several values of Δ; acceptance rises as T increases.

Metropolis numeric example — vary Δ ($T = 2$)

Δ	Acceptance $e^{-\Delta/T}$	Interpretation
0.5	$e^{-0.25} \approx 0.779$	likely accept
1	$e^{-0.5} \approx 0.607$	often accept
2	$e^{-1} \approx 0.368$	coin-flip
3	$e^{-1.5} \approx 0.223$	sometimes accept
5	$e^{-2.5} \approx 0.082$	rarely accept

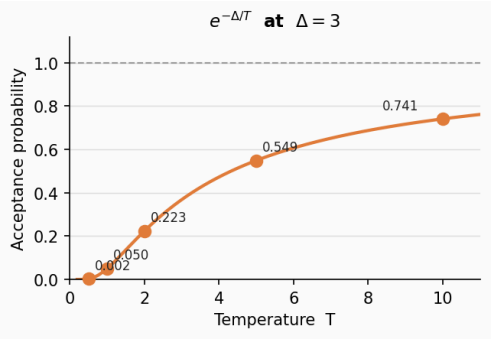
Takeaway. Fixed $T = 2$: acceptance falls as Δ grows. If $\Delta \leq 0$, acceptance is always 1.



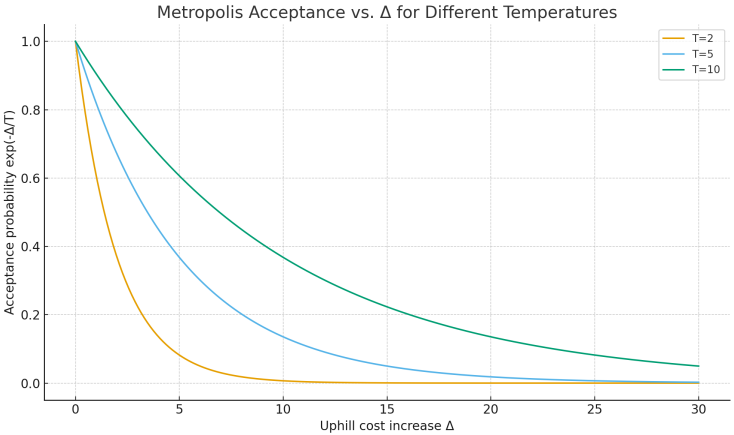
Metropolis numeric example — vary T ($\Delta = 3$)

T	Acceptance $e^{-\Delta/T}$	Interpretation
0.5	$e^{-6} \approx 0.00248$	almost never
1	$e^{-3} \approx 0.0498$	rarely accept
2	$e^{-1.5} \approx 0.223$	sometimes accept
5	$e^{-0.6} \approx 0.549$	often accept
10	$e^{-0.3} \approx 0.741$	likely accept

Takeaway. Fixed $\Delta = 3$: acceptance rises as T increases. If $\Delta \leq 0$, acceptance is 1 for any T.



Metropolis vs. Δ



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SA pseudocode (1/2) — signature & initialisation

```python
from math import exp, log

def simulated_annealing(
    x0,
    N,              # N(x)        -> iterable of moves
    delta,          # delta(m,x) -> O(1) cost change
    f,              # objective function
    *,
    accept_ratio = 0.8,   # target uphill-acceptance for T0
    T0           = None,  # calibrated automatically if None
    alpha        = 0.95,  # geometric cooling factor
    L            = 100,   # iterations per temperature stage
    max_stages   = 200,   # number of cooling stages
    stop         = None,  # stop(x, best, T) -> bool
    rng          = None,  # random.Random instance or None
):
    x    = x0
    best = copy(x)

    if T0 is None:                          # --- calibrate T0 ---
        uphill = sample_uphill_deltas(
                     x, N, delta, M=200, rng=rng)
        T0 = -mean(uphill) / log(accept_ratio)   # proxy formula
    T = T0
```

SA pseudocode (2/2) — main loop

```
    for _ in range(max_stages):             # --- temperature loop ---
        n_accepted = 0
        for _ in range(L):                  # inner Markov chain at T
            m = pivot(N(x), rng)            # pick a random move or using other pivor
            d = delta(m, x)                 # O(1) cost change

            if d <= 0 or rng.random() < exp(-d / T):
                x           = apply(m, x)   # accept move
                n_accepted += 1
                if f(x) < f(best):
                    best = copy(x)          # track global best

        T *= alpha                          # geometric cooling
        if stop and stop(x, best, T):       # optional early exit
            break

    return best
```

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Cooling: why and how it works

- At fixed temperature T, an uphill move of cost $\Delta > 0$ is accepted with probability $\exp(-\Delta/T)$.
- **Goal of cooling:** start *exploratory* (high $T \Rightarrow$ many uphill accepts) then *intensify* (low $T \Rightarrow$ near-greedy).
- We run stages $k = 0, 1, \ldots, K$ at temperatures T_k; each stage has L inner iterations.
- Early stages target a high uphill-acceptance ratio (e.g. 0.7–0.9); late stages aim for ≈ 0.

In the code:

```
for _ in range(max_stages):      # outer: temperature stages (K total)
    for _ in range(L):           # inner: Markov chain at fixed T
        ...                      # Metropolis accept/reject
    T *= alpha                   # <- geometric cooling each stage
```

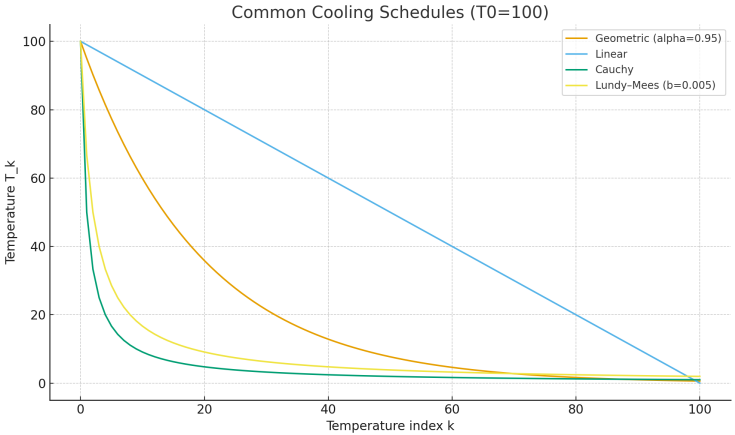
Cooling schedules — mathematically

Given T_0, $T_{\min}$, and number of stages K:

Schedule	Definition	Parameter from $T_0, T_{\min}, K$
Geometric	$T_k = \alpha^k T_0$	$\alpha = \left(\frac{T_{\min}}{T_0}\right)^{1/K}$
Linear	$T_k = T_0 - \beta k$	$\beta = \frac{T_0 - T_{\min}}{K}$
Cauchy	$T_k = \frac{T_0}{1 + c\,k}$	$c = \frac{1}{K}\left(\frac{T_0}{T_{\min}} - 1\right)$
Lundy–Mees	$T_{k+1} = \frac{T_k}{1 + b\,T_k}$	$b = \frac{1/T_{\min} - 1/T_0}{K}$

When to use: Geometric is a robust default. Cauchy/Lundy–Mees are preferable if you want a long, slow late phase. Linear is simple but can undercool unless K is tuned tightly.

Cooling laws — visual



Rule of thumb. Geometric is robust; Cauchy/Lundy–Mees cools fast early with a long tail.

Calibrating T_0 — rationale

The Metropolis criterion accepts an uphill move of size $\Delta > 0$ with probability $e^{-\Delta/T}$. The *mean* acceptance over all uphill moves is our lever:

T_0 too low	T_0 just right	T_0 too high
$e^{-\Delta/T} \approx 0$	$\mathbb{E}[e^{-\Delta/T}] \approx \rho_0$	$e^{-\Delta/T} \approx 1$
No uphill moves accepted. SA reduces to greedy LS immediately — gets trapped in the first local optimum it hits.	Most improvements *and* a controlled fraction of uphill moves accepted. SA can explore broadly before intensifying.	Nearly all moves accepted regardless of cost. SA performs a near-random walk — wastes most of its budget without improving.

Target acceptance ratio ρ_0: typically 0.7–0.9. A value of 0.8 means "at the start, accept 80% of uphill moves" — enough freedom to escape local optima, enough direction to make progress.

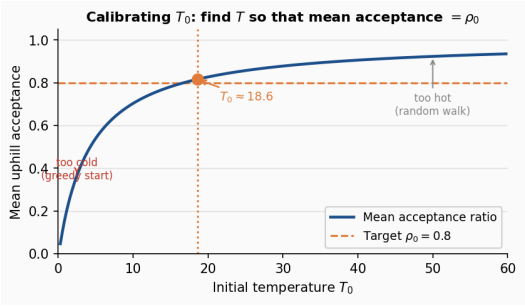
T_0 calibration — procedure & proxy formula

Goal. Find T_0 so that $\frac{1}{M}\sum_{\Delta_m>0} e^{-\Delta_m/T_0} = \rho_0$.

Step-by-step.

1. Sample $M \sim 200$ random moves from x_0.
2. Keep uphill ones ($\Delta > 0$); compute mean $\bar{\Delta}$.
3. **Proxy** (exponential-Δ assumption):

$$\boxed{T_0 \approx \frac{-\bar{\Delta}}{\ln \rho_0}}$$



Mean uphill acceptance vs. T. The orange dashed line marks $\rho_0 = 0.8$; the intersection gives the calibrated T_0. Too cold (left) $\Rightarrow$ greedy start. Too hot (right) $\Rightarrow$ random walk.

Note. If acceptance $< 10\%$ in stage 1, double T_0 and retry. If no uphill-Δ sample is available, manual tuning of `T0` is necessary (or run a short pilot to estimate it).

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Representation & neighbourhood

- Choose a representation enabling cheap local updates.
- Neighbourhood examples:
  - TSP: 2-opt, 3-opt, ...
- Candidate lists & filters: see local search!

Δ-evaluation & caching

- Maintain caches so move deltas are $O(1)$ or local-time.
- Goal: Fast Δ-eval $\Rightarrow$ cheap Metropolis tests $\Rightarrow$ allow more steps in less time.

Acceptance & constraints

Two philosophies:

1. **Feasible-only SA**: generate only admissible moves (simpler; can freeze early).
2. **Penalty SA**: minimise $F(x) = f(x) + \sum_k \lambda_k \, \mathrm{viol}_k(x)$ and allow temporary infeasibility.

Adaptive penalties: increase λ_k when constraint k is violated; decrease when consistently satisfied.

Stopping criteria

Four common stopping rules — use one or combine them:

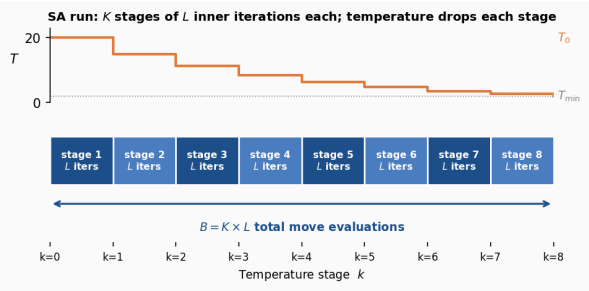
Criterion	When it fires	Typical use
Max stages K	After exactly K temperature stages.	Always use as a hard cap. Simplest; derived from budget $B = K \cdot L$.
Time budget	Wall-clock time $\geq t_{\max}$.	Production / competitions where runtime matters more than iterations.
No-improvement	Best solution unchanged for W consecutive stages.	Good "exploitation tail" guard; set $W \sim 5\%$ of K.
Frozen acceptance	Acceptance ratio $< \epsilon$ for several stages.	Most principled: SA has become greedy, so further cooling wastes budget.

```
        if stop and stop(x, best, T):   # user-supplied predicate
            break                       # fires any of the above rules
```

Budgeting: how B, L, K, and α connect

The four quantities are coupled:

1. Fix the total budget B (move evaluations or wall time).
2. Choose chain length L per stage. $\Rightarrow$ number of stages $K = \lfloor B/L \rfloor$.
3. Choose $T_{\min}$ (near-greedy endpoint). $\Rightarrow$ cooling factor $\alpha = \left(T_{\min}/T_0\right)^{1/K}$.



Each coloured block is one temperature stage (L inner iterations). Temperature steps down after every stage. Total budget $B = K \times L$.

Key trade-off.

- Large $L \Rightarrow$ fewer stages $\Rightarrow$ faster cooling (smaller α).
- Small $L \Rightarrow$ more stages $\Rightarrow$ noisier acceptance estimates per stage.

Typical choices: $L = c \cdot n$ (e.g. $c \in [5, 20]$); $T_{\min}$ small enough that uphill acceptance is effectively zero.

If your budget is time (seconds), not iterations

- Measure iteration rate r (iterations/s) with a short warm-up at representative T:

$$r \approx \frac{\text{iterations in warmup}}{\text{warmup time (s)}}$$

- Convert to an iteration budget:

$$\boxed{B \approx r \times \text{time budget (s)}}$$

- Continue as before: $K \approx \lfloor B/L \rfloor$, $\alpha = (T_{\min}/T_0)^{1/K}$.

Working example

- Total iteration budget: $B = 10^6$.
- Problem size: $n = 500$. Choose $L = 10n = 5{,}000$ iterations/stage.
- Stages: $K \approx \dfrac{B}{L} = \dfrac{10^6}{5{,}000} = 200$.
- Calibrated $T_0 = 15$. Target end temperature $T_{\min} = 0.02$.
- Cooling factor:

$$\alpha \;=\; \left(\frac{0.02}{15}\right)^{1/200} \;\approx\; 0.967.$$

Practical picks for B, L, K, α

- **B**: set by time/memory constraints; prefer round numbers (e.g., 10^6, 2×10^6).
- **L**: start with $L = c \cdot n$ (try $c \in [5, 20]$); increase if acceptance estimates are too noisy.
- **K**: $K \approx B/L$. If $K < 50$, consider larger B or smaller L to smooth cooling.
- $\boldsymbol{\alpha}$: compute from $T_0, T_{\min}, K$. If the run ends too "hot," decrease $T_{\min}$ or increase K.
- *Heuristic target:* pick $T_{\min}$ so that $e^{-\Delta/T_{\min}} \lesssim 10^{-3}$ for a *typical* uphill Δ.

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TSP: problem instance

5-city symmetric TSP. Cities $\{1, 2, 3, 4, 5\}$; integer distance matrix:

	1	2	3	4	5
1	–	10	8	9	7
2	10	–	6	4	9
3	8	6	–	5	11
4	9	4	5	–	8
5	7	9	11	8	–

Initial tour: $\pi_0 = (1, 2, 3, 4, 5)$, cost $= d_{12} + d_{23} + d_{34} + d_{45} + d_{51} = 10 + 6 + 5 + 8 + 7 = \mathbf{36}$.

Representation: permutation vector π. **Moves:** 2-opt (and generalisations: 3-opt, Or-opt).

2-opt neighbourhood & Δ-evaluation

Move. Choose two *non-adjacent* tour edges (π_i, π_{i+1}) and (π_j, π_{j+1}), remove them, and reconnect by reversing the segment $\pi_{i+1} \cdots \pi_j$.

$O(1)$ **cost change** — only 4 table look-ups:

$$\Delta \;=\; \underbrace{d(\pi_i, \pi_j) + d(\pi_{i+1}, \pi_{j+1})}_{\text{new edges added}} \;-\; \underbrace{d(\pi_i, \pi_{i+1}) + d(\pi_j, \pi_{j+1})}_{\text{old edges removed}}.$$

Worked example — swap edges $(1, 2)$ and $(3, 4)$ in $\pi_0 = (1, 2, 3, 4, 5)$:

$$\Delta = \big[d_{13} + d_{24}\big] - \big[d_{12} + d_{34}\big] = \big[8 + 4\big] - \big[10 + 5\big] = -\mathbf{3}.$$

$\Delta < 0 \Rightarrow$ always accepted. New tour: $(1, 3, 2, 4, 5)$, cost $= 33$.

All 2-opt moves from $\pi_0 = (1, 2, 3, 4, 5)$, cost $= 36$

$n = 5 \Rightarrow \frac{n(n-3)}{2} = 5$ non-adjacent edge pairs:

Edges removed	New tour	Δ	$p\,(T{=}5)$	$p\,(T{=}0.5)$
$(1,2)\,\&\,(3,4)$	$(1,3,2,4,5)$	-3	1	1
$(1,2)\,\&\,(4,5)$	$(1,4,3,2,5)$	0	1	1
$(2,3)\,\&\,(4,5)$	$(1,2,4,3,5)$	$+1$	$e^{-0.2} \approx 0.819$	$e^{-2} \approx 0.135$
$(2,3)\,\&\,(5,1)$	$(1,2,5,4,3)$	$+4$	$e^{-0.8} \approx 0.449$	$e^{-8} \approx 3.4 \times 10^{-4}$
$(3,4)\,\&\,(5,1)$	$(1,2,3,5,4)$	$+8$	$e^{-1.6} \approx 0.202$	$e^{-16} \approx 1.1 \times 10^{-7}$

Takeaway. At warm $T{=}5$, even a barrier of $+8$ passes $\sim$20% of the time. At cold $T{=}0.5$, only improvements and neutral moves are accepted in practice.

Mini SA trace on the 5-city TSP

$\pi_0 = (1, 2, 3, 4, 5)$, cost $= 36$; steps 1–3 at $T = 5$, step 4 at $T = 0.5$.

Step	Move (2-opt)	Δ	$e^{-\Delta/T}$ vs. rng	Decision
1	remove $(1, 2)$ & $(3, 4)$	-3	—	**Accept** (downhill)
	$\Rightarrow \pi = (1, 3, 2, 4, 5)$, cost $= 33$			
2	remove $(3, 2)$ & $(4, 5)$	0	—	**Accept** (neutral)
	$\Rightarrow \pi = (1, 3, 4, 2, 5)$, cost $= 33$			
3	remove $(1, 3)$ & $(4, 2)$	$+3$	$0.549 > 0.42$	**Accept** (uphill!)
	$\Rightarrow \pi = (1, 4, 3, 2, 5)$, cost $= 36$			
4	remove $(3, 2)$ & $(5, 1)$	$+8$	$1.1 \times 10^{-7} < 0.72$	**Reject**
	(now $T = 0.5$; uphill barrier too large)			

Step 3 shows SA's key feature: an uphill move is accepted at high T enabling escape from a local optimum. Step 4 shows how cooling "freezes out" such moves.

TSP: calibrating T_0

Procedure. Sample M random 2-opt moves; keep uphill Δ-values; solve

$$\frac{1}{M} \sum_{m:\, \Delta_m > 0} e^{-\Delta_m / T_0} = \rho_0 .$$

Proxy formula (accurate when the uphill Δ-distribution is roughly exponential):

$$T_0 \approx \frac{-\bar{\Delta}}{\ln \rho_0} .$$

Our 5-city instance. Uphill Δ-values: $\{1, 4, 8\}$, mean $\bar{\Delta} \approx 4.3$.

$$T_0 \approx \frac{4.3}{0.223} \approx 19.3.$$

Δ	$e^{-\Delta/19.3}$
1	$e^{-0.052} \approx 0.949$
4	$e^{-0.207} \approx 0.813$
8	$e^{-0.415} \approx 0.660$
mean	$\approx 0.807 \approx \rho_0 = 0.8$ ✓

For real instances, sample hundreds of moves for a reliable $\bar{\Delta}$.

TSP: budget & schedule ($n = 20$)

- Problem: $n = 20$ cities; 2-opt neighbourhood size $= \frac{n(n-3)}{2} = 170$ moves.
- **Iteration budget:** $B = 200{,}000$.
- **Inner-loop length:** $L = 10n = 200$ iterations/stage.
- **Number of stages:** $K = \dfrac{B}{L} = \dfrac{200{,}000}{200} = 1{,}000$.
- **Calibrated** $T_0 = 50$; target end temperature $T_{\min} = 0.05$.
- **Cooling factor:**

$$\alpha = \left(\frac{T_{\min}}{T_0}\right)^{1/K} = \left(\frac{0.05}{50}\right)^{1/1000} = 0.001^{0.001} \approx 0.9931.$$

- **Sanity check.** At $T_{\min} = 0.05$, a typical uphill $\Delta = 4$:

$$e^{-4/0.05} = e^{-80} \approx 0 \;\Rightarrow\; \text{greedy end-game.}$$